Forward-Looking Expectations and the ICEV Sale Ban: Mathematical Documentation of Model Extensions

Abstract

NEED TO FIX THE BAN IMPLEMENTATION The base model (as described in the manuscript "Driving in the wrong direction? Modelling policy mixes for EV adoption") gives agents and firms *naive, permanent-policy* expectations: whatever price or policy applies today is assumed, in every lifecycle-cost calculation, to hold forever. This note documents two extensions built on top of that baseline. First, a *forward-looking expectations* mechanism that replaces the naive constant-price assumption with a discounted sum over the model's actual, fully-known future price/policy path, applied consistently to both consumers and firms, and used in the `package/variable_carbon_price` experiment to compare naive and forward-looking responses to a rising carbon price. Second, an *ICEV sale ban*, a discrete future policy event (rather than a continuous price signal), together with a matching notion of anticipation for forward-looking firms, used in the `package/car_ban` experiment to compare ban years 2030–2035. Both extensions are implemented so that, with the new flags left at their default (off) values, the model's behaviour is byte-for-byte identical to the pre-existing baseline.

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1 Baseline: naive, permanent-policy expectations

1.1 The lifecycle-cost term

For a vehicle a of transport type ICE or EV, owned or considered by individual i, the model's per-period utility includes a term that penalises the discounted lifecycle fuel/emissions cost of operating that vehicle forever. Writing ω for the vehicle's fuel/energy efficiency, δ for its efficiency depreciation rate, L for its current age (in months), r for the individual's/firm's discount rate, D_i for individual i's driving distance, c for the per-unit fuel or electricity cost, e for the per-unit emissions intensity, and γ_i for individual i's emissions willingness-to-pay, the existing (pre-extension) formula for this term is

$$\mathrm{LCC}_{\mathrm{extra}} = D_i \cdot \frac{(1+r)(1-\delta)\left(c+\gamma_i e\right)}{\omega\,(1-\delta)^L\left(r-\delta-r\delta\right)}. \tag{1}$$

This is exactly the closed-form expression implemented (identically) in three places in the code: `socialNetworkUsers.py`'s `generate_utilities_current`, `vectorised_calculate_utility_second_hand_cars` and `vectorised_calculate_utility_new_cars` (the utility a consumer gets from a vehicle), and `firm.py`'s `calc_optimal_price_cars` / `calc_utility` (a firm's own valuation of a vehicle design, used to set prices and choose what to research/produce).

1.2 Where Eq. (1) comes from

Equation (1) is the closed form of an infinite discounted sum under two assumptions: (i) the per-unit cost/emissions level $z \equiv c+\gamma_i e$ is constant forever, and (ii) the vehicle's fuel efficiency degrades geometrically, $\omega_{L+k} = \omega\,(1-\delta)^k$, k periods after the current age L. The present value, as seen at the vehicle's current age L, of all future per-distance operating costs is

$$C(L) = \sum_{k=0}^{\infty} \frac{1}{(1+r)^k} \cdot \frac{z}{\omega\,(1-\delta)^{L+k}} = \frac{z}{\omega\,(1-\delta)^L} \sum_{k=0}^{\infty} q^k, \qquad q \equiv \frac{1}{(1+r)(1-\delta)}. \tag{2}$$

Provided $r > \delta/(1-\delta)$ (equivalently $q < 1$, the convergence condition already checked in `controller.py`), the geometric series sums to $1/(1-q)$, and a short calculation gives

$$\frac{1}{1-q} = \frac{1}{1-\dfrac{1}{(1+r)(1-\delta)}} = \frac{(1+r)(1-\delta)}{(1+r)(1-\delta)-1} = \frac{(1+r)(1-\delta)}{r-\delta-r\delta}, \tag{3}$$

using $(1+r)(1-\delta)-1 = r-\delta-r\delta$. Substituting (3) into (2) and multiplying by D_i reproduces Eq. (1) exactly. This is the calculation this document refers to as the *naive/permanent* assumption: it correctly discounts the future, but it assumes the price/emissions level z observed *today* will never change — so a rising carbon price, for instance, is invisible to it until the month it actually arrives.

2 Forward-looking expectations

2.1 Generalising the geometric series

The paper itself notes this limitation directly (Section 4.1): "we disregard the role of expectations regarding policy stringency ... with agents assuming that policies will be permanent." The fix does not require abandoning Eq. (2)'s structure, only its assumption that z is constant. Write z_t for the per-unit cost or emissions level at absolute simulation month t (known for the whole horizon in advance, since it is simply the pre-computed price/carbon-price/subsidy path already built by `controller.py`). Replacing the constant z in (2) by the actual path z_{t+k} gives the forward-looking present-value index

$$\text{Index}(t) \;\equiv\; \sum_{k=0}^{T_{\max}-t} z_{t+k}\, q^k, \qquad q = \frac{1}{(1+r)(1-\delta)}, \tag{4}$$

where $T_{\max}$ is the last simulated month. When $z_t = z$ for all t, (4) collapses to $z/(1-q)$, i.e. exactly the naive case — so Eq. (4) is a strict generalisation of Eq. (2), not a different model.

2.2 Backward recursion

Rather than evaluating the sum in (4) from scratch at every t (an $O(T_{\max}^2)$ computation, and one that would have to be repeated per agent if it were not for the separation argument in Section 2.3), it is computed once, for the whole horizon, via the backward recursion

$$\text{Index}(t) = z_t + q\,\text{Index}(t+1), \qquad \text{Index}(T_{\max}) = \frac{z_{T_{\max}}}{1-q}, \tag{5}$$

which is easily verified by unrolling (4) one step: $\sum_{k=0}^{K} z_{t+k} q^k = z_t + q \sum_{k=0}^{K-1} z_{t+1+k} q^k$. The boundary condition treats the last known value $z_{T_{\max}}$ as persisting forever beyond the modelled horizon — the same perpetuity assumption the naive model already makes everywhere, just deferred to "after the simulation stops knowing anything" rather than applied from today onward. This is implemented as `Controller.compute_discounted_indices()` / `Controller._discounted_index()`, a single $O(T_{\max})$ pass computed once per model run (or once per future-period re-run when reusing a calibration, see `setup_continued_run_future`), independent of the number of agents or vehicles.

2.3 Why cost and emissions must be indexed separately

The naive formula (1) combines cost and emissions *before* discounting: $z = c + \gamma_i e$. This is harmless in the naive case because γ_i, although agent-specific, is just a constant multiplier sitting outside an otherwise-agent-independent sum. It stops being harmless once c_t and e_t are allowed to vary independently over time (as they do once carbon price, gasoline price and grid-decarbonisation paths differ): γ_i cannot be pulled inside a sum that is shared across all agents if the combination $c_t + \gamma_i e_t$ is what gets discounted, since that would require recomputing the whole discounted sum once per distinct γ_i value. Instead, the implementation keeps *two* indices,

$$\text{Index}_{\text{cost}}(t) = \sum_k c_{t+k}\, q^k, \qquad \text{Index}_{\text{emissions}}(t) = \sum_k e_{t+k}\, q^k, \tag{6}$$

each computed once (globally, not per-agent, since r and δ are global scalars shared by all ICE, resp. all EV, vehicles), and combines them with the agent-specific γ_i *outside* the sum, at the point of use:

$$\text{LCC}^{\text{FL}}_{\text{extra}} = D_i \cdot \frac{\text{Index}_{\text{cost}}(t) + \gamma_i\, \text{Index}_{\text{emissions}}(t)}{\omega\,(1-\delta)^L}. \tag{7}$$

This is the forward-looking replacement for Eq. (1). Note the age-decay factor $\omega\,(1-\delta)^L$ is unchanged: it depends only on the vehicle's efficiency and current age, not on the price/emissions path, so it factors out of the sum in (2) identically in both the naive and forward-looking cases. For newly-designed cars (age $L = 0$, as evaluated by firms in Section 2.5), this factor is simply ω.

2.4 The four price/emissions paths

Four scalar paths are discounted via (5), one pair (cost, emissions) for each of the two energy carriers:

$$z^{\text{gas}}_{\text{cost}}(t) = \text{GasPrice}(t) + \text{CarbonPrice}(t) \cdot \bar{e}_{\text{gas}}, \qquad z^{\text{gas}}_{\text{em}}(t) = \bar{e}_{\text{gas}} \quad \text{(constant: ICE tailpipe intensity doe} \tag{8}$$

$$z^{\text{elec}}_{\text{cost}}(t) = \text{ElecPrice}(t)\,\big(1 - s(t)\big) + \text{CarbonPrice}(t) \cdot \bar{e}_{\text{grid}}(t), \quad z^{\text{elec}}_{\text{em}}(t) = \bar{e}_{\text{grid}}(t), \tag{9}$$

where $s(t)$ is the electricity price subsidy proportion and $\bar{e}_{\text{grid}}(t)$ is the (time-varying, decarbonising) grid emissions intensity. These are exactly `gas_price_california_vec`, `carbon_price_time_series`, `gas_emissions_intensity`, `electricity_price_vec`, `electricity_price_subsidy_time_series` and `electricity_emissions_intensity_vec` already built by `controller.py`'s existing calibration/scenario/policy machinery — **no new exogenous path had to be built**; only the mechanism that reads it changed. Each of the four resulting index arrays ($\text{Index}^{\text{gas}}_{\text{cost}}$, $\text{Index}^{\text{gas}}_{\text{em}}$, $\text{Index}^{\text{elec}}_{\text{cost}}$, $\text{Index}^{\text{elec}}_{\text{em}}$) is looked up at the current month and written onto every car of the matching transport type as `car.cost_index` / `car.emissions_index`, exactly mirroring how `car.fuel_cost_c` / `car.e_t` are already refreshed every timestep.

2.5 Firms

The identical substitution (Eq. (1) → Eq. (7)) is applied on the firm side, in `Firm._lifecycle_cost_term_firm`, used both when a firm prices a car for sale (`calc_optimal_price_cars`) and when it values a candidate design during R&D (`calc_utility`, called from `calc_utility_cars_segments` and `set_utility_and_profit_grid_production`). Since a firm always prices/evaluates a freshly designed car, its age is $L = 0$ and the age-decay factor is simply ω (i.e. `age_factor=1` in the code, verified to be bit-for-bit equivalent to omitting the factor entirely).

2.6 A single switch, two exact regimes

Everywhere above, the choice between Eq. (1) and Eq. (7) is governed by one boolean, `forward_looking_expectati` read once per run (default `False`) and shared by every consumer and every firm in that run. When it is `False`, the code path is Eq. (1), expressed with the exact same operations/operand order as the original code — so existing configurations that never set this flag get bit-for-bit identical output. When it is `True`, every consumer and firm uses Eq. (7), driven by the same four index arrays.

2.7 Application: the `variable_carbon_price` experiment

`package/variable_carbon_price` exploits Eq. (7) directly: the carbon-price path CarbonPrice(t) is set to $\sim$ 10 different linearly-rising ramps (reaching a range of end-prices at the end of the policy horizon) and to 5 flat carbon taxes at correspondingly *half* the ramps' end-prices (so that a flat tax and its reference ramp have the same time-averaged carbon price — a linear ramp from 0 to X averages $X/2$, so comparing it to a flat tax of X would overstate how much cheaper the ramp looks). Every scenario is run under both `forward_looking_expectations` $\in \{\texttt{False}, \texttt{True}\}$. Under naive expectations, a rising carbon price is met reactively, one month at a time, as though each month's price were permanent; under forward-looking expectations, Eq. (4) means agents and firms price in the *entire future ramp* from the start, so EV adoption should shift earlier and be less sensitive to the exact shape of the ramp, relative to the naive case. This is exactly the comparison the experiment's naive-vs-FL panels are designed to surface.

3 ICEV sale ban

3.1 Why this needs a different mechanism

The carbon-price mechanism of Section 2 is a *continuous cost signal*: at every month, every vehicle has a well-defined operating cost, and forward-looking agents simply discount the known future path of that cost. A sale ban is not a cost signal at all — it is a discrete, mandatory constraint on what firms are legally permitted to *sell*. Eq. (7) therefore does not apply here; the ban is instead expressed with indicator functions governing which vehicle designs a firm may research or produce at each month.

3.2 Ban timing

Let T_{ban} be the absolute simulation month at which the ban takes legal effect (configured as `ICE_ban_time`, using the same "months after burn-in ends" convention as the pre-existing `ev_research_start_time` and let Λ be an anticipation lead time in months (`ICE_ban_anticipation_lead`). Define the *research wind-down month*

$$T_{\text{wind}} = T_{\text{ban}} - \Lambda \cdot \mathbb{1}\big[\texttt{forward\_looking\_expectations}\big]. \tag{10}$$

For a naive firm, $T_{\text{wind}} = T_{\text{ban}}$: it keeps researching and selling ICE cars right up to the legal deadline and is, in effect, surprised by the ban. For a forward-looking firm, $T_{\text{wind}} < T_{\text{ban}}$: it anticipates the ban and stops *investing in new ICE designs* Λ months early, while continuing to legally sell already-existing ICE models until T_{ban} — mirroring the real-world logic that a firm has no reason to voluntarily withdraw an already-profitable, still-legal product ahead of when it is actually forced to.

3.3 Firm-level indicator functions

Each firm f carries two boolean state variables, $\mathbb{1}^{\text{res}}_{\text{ICE}}(f,t)$ and $\mathbb{1}^{\text{prod}}_{\text{ICE}}(f,t)$ (`ice_research_bool`, `ice_production_bool` in the code), both initialised to 1 (i.e. ICE research/production behave exactly as in the un-extended model whenever no ban is configured) and updated by the controller at the appropriate months:

$$\mathbb{1}^{\text{res}}_{\text{ICE}}(f,t) = \begin{cases} 1 & t < T_{\text{wind}} \\ 0 & t \geq T_{\text{wind}} \end{cases} \qquad \mathbb{1}^{\text{prod}}_{\text{ICE}}(f,t) = \begin{cases} 1 & t < T_{\text{ban}} \\ 0 & t \geq T_{\text{ban}} \end{cases} \tag{11}$$

These gate two sets used elsewhere in the firm's decision process: the set of technologies eligible for R&D that month,

$$\mathcal{N}_{\text{research}}(f,t) = \Big(\mathcal{N}_{\text{ICE}}(f,t) \cdot \mathbb{1}^{\text{res}}_{\text{ICE}}(f,t)\Big) \cup \Big(\mathcal{N}_{\text{EV}}(f,t) \cdot \mathbb{1}_{\text{EV}}\Big), \tag{12}$$

and the memory pool eligible for production/pricing that month,

$$\mathcal{M}_{\text{production}}(f,t) = \Big(\mathcal{M}_{\text{ICE}}(f) \cdot \mathbb{1}^{\text{prod}}_{\text{ICE}}(f,t)\Big) \cup \Big(\mathcal{M}_{\text{EV}}(f) \cdot \mathbb{1}_{\text{EV}}\Big), \tag{13}$$

where $\mathbb{1}_{\text{EV}}$ is the pre-existing (unmodified) `ev_research_bool`/`ev_production_bool` switch. When $\mathbb{1}^{\text{res}}_{\text{ICE}} = \mathbb{1}^{\text{prod}}_{\text{ICE}} = 1$ for all t (i.e. no ban configured), Eqs. (12)–(13) reduce exactly to the original, unconditional set expressions — so, as with the forward-looking mechanism, an unconfigured ban leaves existing behaviour untouched.

3.4 Immediate market exit at T_{ban}

Eq. (13) alone would only prevent ICE from being *re-selected* the next time a firm happens to re-price its lineup (a stochastic event, governed by `prob_change_production`, that does not necessarily coincide with T_{ban}). Since the experiment requires an exact, comparable cutoff across ban years, the set of cars actually on sale is additionally filtered, exactly at T_{ban}, regardless of whether a re-pricing event fires that month:

$$\text{CarsOnSale}(f, T_{\text{ban}}) = \text{CarsOnSale}(f, T_{\text{ban}}^{-}) \setminus \{\, c : \text{transportType}(c) = \text{ICE} \,\}. \tag{14}$$

This guarantees that no new ICE car is available for purchase from T_{ban} onward, in every scenario, independent of the stochastic re-pricing cadence.

3.5 What is *not* modelled

Two things are deliberately left untouched. First, the second-hand market: existing ICE vehicles can continue to be owned, driven and resold after the ban; only the sale of *new* ICE vehicles by firms is restricted. Second, consumer-side anticipation of the ban itself: the only realistic channel for a consumer to anticipate a future sale ban — a lower expected resale value for an ICE car bought shortly before the ban, since the resale market it will eventually be sold into is smaller — would require vehicle resale-value modelling the ABM does not currently have, and is left for future work. Consumers in this experiment still use whichever of Eq. (1)/(7) corresponds to the run's `forward_looking_expectations` flag for ordinary fuel/carbon-price discounting; the only *ban-specific* anticipation implemented is on the firm side (Eq. (10)), which then reaches consumers indirectly, through whichever vehicles firms actually choose to put on sale.

3.6 Application: the `car_ban` experiment

`package/car_ban` sweeps T_{ban} over calendar years 2030–2035 (converted to absolute months via the same reference point the existing `controller.t_2030` constant uses, i.e. the future period begins at calendar year 2024), each under both `forward_looking_expectations` $\in$ {`False`, `True`} with a fixed anticipation lead Λ (2 years by default), plus a no-ban (BAU) baseline. Comparing the naive and forward-looking columns of each metric's figure shows what a firm's anticipation of the ban — as opposed to the ban itself, which both regimes experience identically at T_{ban} — changes about the transition.

4 Summary of new parameters

Parameter	Default	Meaning
`forward_looking_expectations`	`False`	Eq. (1) vs. Eq. (7)
`ICE_ban_time`	`None` (never)	T_{ban}, Eq. (11)
`ICE_ban_anticipation_lead`	0	Λ, Eq. (10)

Table 1: New backward-compatible controller parameters. All three default to values that reproduce the pre-existing model exactly.